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0.2-1 | Project description =====
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Design of embedded pole foundation for the flagpole design example in Appendix A of NAAMM/FP 1001-07, “Guide Specifications for Design of Metal Flagpoles” Embedded pole foundation design is per 2024 IBC Eq 6-1 and Table 18-I-A. Soil properities are per Table 1806.2 in the 2024 IBC.

0.2-2 | Design input

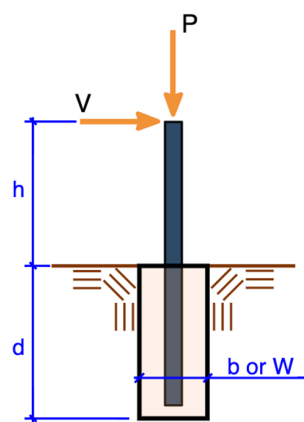


Fig. 2.1 - Calculation Diagram

Table 2.1: Design input

variable	value	[value]	description
Mbase	24.84 ft-kip	33.67 mkN	moment at base of flagpole
P	24.84 kips	110.47 kN	horizontal load
b	24.00 inch	60.96 cm	width of concrete drilled pier
PFP	200.00 p_cf	31.42 kN_m3	allowable lateral bearing pressure - sandy gravel
h	1.50 ft	0.46 m	height
d	10.00 ft	3.05 m	initial guess for embedment depth

Table 2.2: Load embedment functions (pole_embed.py)

Function	Docstring
Depth_nonconstrained(d, P, h, dia, PFP, tol)	Calculate required pole embedment using Eq. 18-1
Depth_constrained(d, P, h, dia, PFP)	Calculate required pole embedment using Eq. 18-2

0.2-3 | Design Results

Eq. 3.2: Required embed - nonconstrained

$\text{depth}_1 = \text{Depth_nonconstrained}(d, P, h, b, \text{PFP}, 2)$

$\text{depth}_1 = 15.52 \text{ ft}$ $[\text{depth}_1] = 4.73 \text{ m}$ | Required embed - nonconstrained

d	h	P	b	PFP
10.00 ft	1.50 ft	24.84 kips	24.00 inch	200.00 p_cf
initial guess for embedment depth	height	horizontal load	width of concrete drilled pier	allowable lateral bearing pressure - sandy gravel
-	-	-	-	-

Eq. 3.3: Required embed - constrained

$\text{depth}_2 = \text{Depth_constrained}(d, P, h, b, \text{PFP})$

$\text{depth}_2 = 7.34 \text{ ft}$ $[\text{depth}_2] = 2.24 \text{ m}$ | Required embed - constrained

d	h	P	b	PFP
10.00 ft	1.50 ft	24.84 kips	24.00 inch	200.00 p_cf
initial guess for embedment depth	height	horizontal load	width of concrete drilled pier	allowable lateral bearing pressure - sandy gravel
-	-	-	-	-

